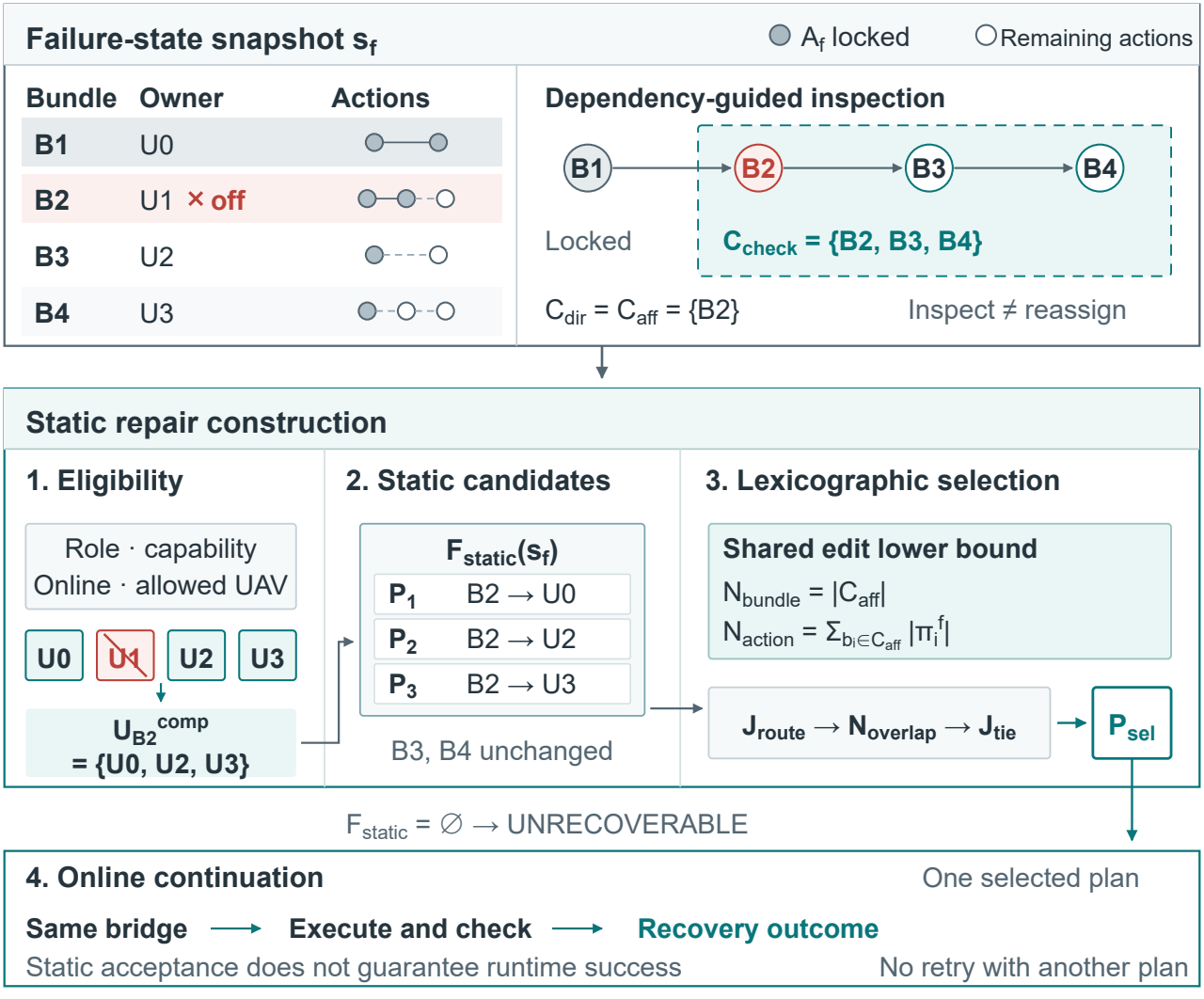

Failure-state snapshot s_f
A_f locked
Remaining actions
Bundle
Owner
Actions
B1
U0
B2
U1 × off
B3
U2
B4
U3
Dependency-guided inspection
B1
B2
B3
B4
Locked
C_{check} = {B2, B3, B4}
C_{dir} = C_{aff} = {B2}
Inspect ≠ reassign
Static repair construction
1. Eligibility
Role · capability
Online · allowed UAV
U0
U1
U2
U3
U_{B2}^{comp}
= {U0, U2, U3}
2. Static candidates
$F_{static}(s_f)$
P_1 B2 → U0
P_2 B2 → U2
P_3 B2 → U3
B3, B4 unchanged
3. Lexicographic selection
Shared edit lower bound
$N_{bundle} = |C_{aff}|$
$N_{action} = \Sigma_{b_i \in C_{aff}} |\pi_i^f|$
J_{route} → $N_{overlap}$ → J_{tie}
P_{sel}
F_{static} = ∅ → UNRECOVERABLE
4. Online continuation
One selected plan
Same bridge → Execute and check → Recovery outcome
Static acceptance does not guarantee runtime success
No retry with another plan